Lesson 14: Raising the Stakes

Scratch

VIDEO GAME CREATION

This week: later waves get genuinely harder - more enemies, moving faster - using the exact wave variable you've had since Lesson 11.

Open your own [YourName]Game project from Lesson 13 - don't start a new one.

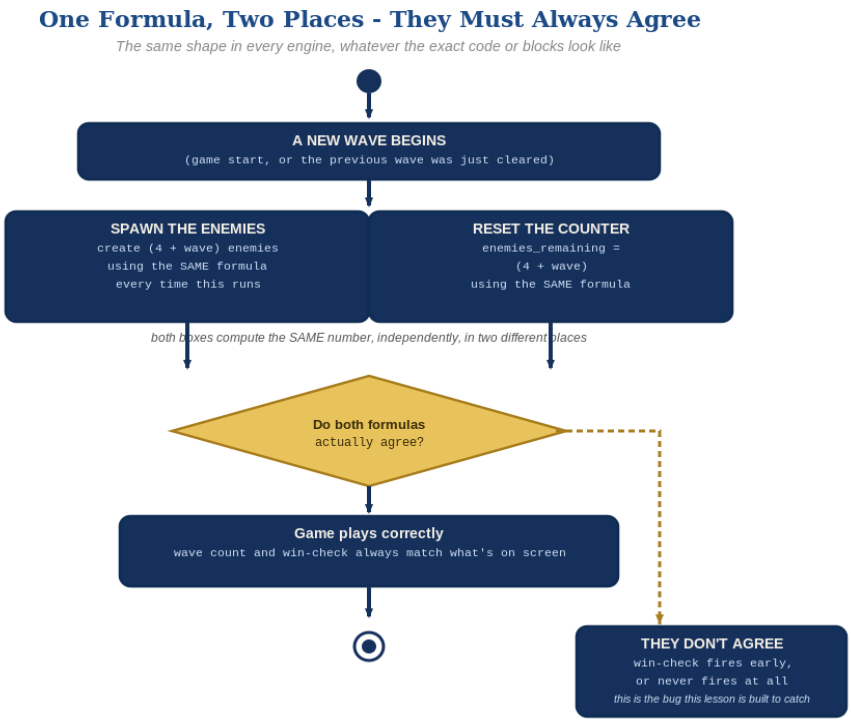
Tip: no new assets this week - difficulty scaling reuses your exact existing enemy sprites and spawn logic.

Setup

1. Open your [YourName]Game project from Lesson 13.
2. No new variables needed for Level 1 - Level 2 will add one (enemy_speed).

Level 1 - Minimal (MVP)

Each wave spawns one more enemy than the last (5, 6, 7). Speed stays the same for now - that's Level 2.



Whether it's two places (most engines) or the loop and the reset living in genuinely different scripts (Godot, GameMaker), the rule is the same: one formula, referenced everywhere.

VIDEO GAME CREATION

Whether the spawn loop and the reset live in two places or two genuinely different scripts, the rule is the same: one formula, referenced everywhere.

3. On the Enemy sprite's "when I receive [spawn enemies]" script, change the fixed "repeat 5" loop into "repeat (4 + wave)" (code below) - wave 1 spawns 5, wave 2 spawns 6, wave 3 spawns 7.
4. Wherever enemies_remaining gets reset to 5 for a new wave (both on GameManager's "when I receive [start game]" script and in the "when I receive [enemy hit]" wave-advance branch), change that to "set enemies_remaining to (4 + wave)" too - these two places must always agree.
5. Click the stage and play through a full round - wave 2 should visibly spawn one more enemy than wave 1, and wave 3 one more than that.

Highlighted lines are new this week; everything else is code you already have.

```
Enemy's "when I receive [spawn enemies]" script (revised):
when I receive [spawn enemies]
repeat (4 + wave)
  create clone of myself

GameManager's "when I receive [start game]" script (revised,
enemies_remaining line only):
set enemies_remaining to (4 + wave)

GameManager's "when I receive [enemy hit]" wave-advance (else) branch
(revised, enemies_remaining line only):
set enemies_remaining to (4 + wave)
```

Level 2 - Complete (Alpha/Beta)

Enemy speed ALSO increases each wave, on top of the growing count. Restarting correctly returns everything to wave 1's original values.

- Add a new variable enemy_speed (for all sprites). In the Enemy's "when I receive [spawn enemies]" script, add "set enemy_speed to (1 + (wave - 1) * 0.5)".
- In the Enemy clone's forever loop, change "change y by -1" to "change y by (enemy_speed * -1)".
- Press R to restart mid-game and confirm both count and speed return correctly to wave 1's values, not wherever wave 3 left them.
- Play a full round and confirm wave 3 is noticeably busier and faster than wave 1, but still fair.

Level 3 - Refined - Choose ONE Stretch Feature

Pick exactly one of the five ideas below. These are intentionally open-ended - a good-faith attempt at the general idea is a complete Level 3 finish, not a fully polished feature.

- Destructible shields/barriers: a few simple blocks placed between the player and the enemies that disappear (or shrink) after absorbing a hit or two - reuses your existing collision-detection and hit-reaction patterns from Lessons 4 and 13, just on a new kind of object.
- A power-up: an item that occasionally drops or spawns, which the player collects to temporarily change how shooting works (faster fire rate, multiple bullets at once, or a brief shield) - reuses your existing timer/cooldown pattern from Lesson 3.
- A bonus enemy: a rare, faster-moving, or oddly-patterned enemy worth extra points, spawned occasionally alongside a normal wave - reuses this week's own speed/count scaling ideas, just for one special case instead of the whole wave.
- Formation-marching movement: instead of each enemy independently drifting down (this course's approach since Lesson 4), have the whole wave move together as a group and occasionally step sideways - a genuinely different movement pattern, inspired by classic arcade shooters.
- A title screen and/or game-over screen: a proper opening screen (press a key to start) and/or a dedicated game-over screen with a final score, instead of just text overlaid on the gameplay - reuses your existing game_active flag and UI tools from Lessons 9-10. Scratch's "switch backdrop" blocks are a natural fit for a simple title/game-over screen.

If Something's Wrong

- Every wave still spawns exactly 5: confirm the loop actually uses "(4 + wave)" and not a leftover hardcoded 5.
- "You Win!" fires early or never: confirm enemies_remaining is reset using the SAME formula in every place it's set, not 5 in one spot and (4 + wave) in another.
- Speed doesn't change between waves: confirm enemy_speed is recalculated inside the Enemy's "when I receive [spawn enemies]" script (not just once at game start) and that the Enemy's forever loop is actually reading enemy_speed, not a leftover -1.

Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas